

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Concept Mapping Questions

Course Code:	CSA0315	Course Title:	Data Structures
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Concept Mapping Questions
Weightage:	10% of CIA	Total Marks:	100
CO1 Statement:	Basics, Data, Data object, Data structure, Abstract Data Types (ADT), realization of ADT in 'C', Primitive and non-primitive data structure, Linear and non-linear data structure Arrays & Recursion, Performance analysis and Measurement: Time and space analysis of algorithms, Average, best and worst-case analysis.		
Student Name:	K.Madhulika	Reg. No.:	192524190
Section / Batch:	Slot D	Date:	

Code of Conduct

I _____K.Madhulika(192524190)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____Madhulika _____

Instructions

- This test contains 3 Concept mapping questions. Total: 100 Marks.
- When a student answers using a concept map, in evaluation the answer should contain the following four requirements: Understanding of concepts, Connections between ideas, Logical structure, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

Section / Topic	Focus	Q Nos	Marks
Basics, Data, Data object, Data structure, Abstract Data Types (ADT), realization of ADT in 'C'	Concept mapping of Basics in Data Structure	1	30
Performance analysis and Measurement: Time and space analysis of algorithms, Average, best and worst-case analysis.	Fibonacci series	2	40
Primitive and non-primitive data structure, Linear and non-linear data structure Arrays	Concept mapping on classifications of data structures.	3	40

Annexure A –Application Based Assessment

1) Write a c program and draw the flow chart as concept map for finding the count of odd and even numbers in an array of n numbers.

2) Write a concept map based flow chart and also write the program to generate the Fibonacci series and sum of the series up to n numbers.

3)A smart city traffic management system handles data related to vehicle movement, road networks, and signal control. Construct a concept map connecting Primitive Data Structures → Non-Primitive Data Structures → Linear → Non-Linear Structures. Include examples such as arrays, linked lists, trees, and graphs, and illustrate their applications in traffic management. Explain the relationships between these structures and justify their classification based on functionality and efficiency.

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding ; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution

		justification			
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

CO1 - AT1

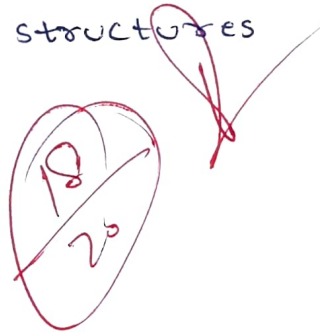
Name : K. Madhulika

Reg no: 192524190

Course code: CSA 0315

Course name: Data structures

Date: 14/7/26



CO1 - AT1

Write a c program to count number of odd and even numbers in an array of numbers.

a. Aim:

To write a c program to count the number of odd and even numbers in an array.

Algorithm:

1. Start
2. Read the number of elements.
3. Read the array elements.
4. Initialize odd=0 and even=0.
5. Check each element.
6. Display the number of odd and even elements.
7. Stop.

Program:

```
#include <stdio.h>
```

```
int main( )
```

```
{
```

```
    int n, i, a[50];
```

```
    int odd=0, even=0;
```

```
    printf("enter number of elements:");
```

```
    scanf("%d", &n);
```

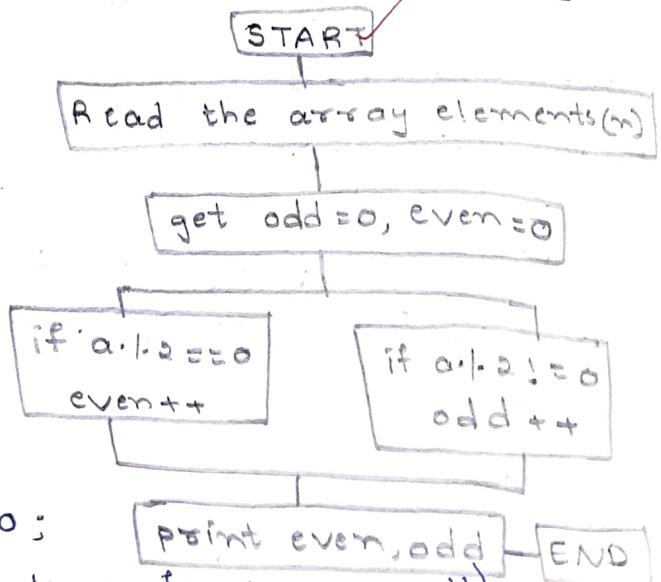
```
    printf("enter array elements: ");
```

```
    for(i=0; i<n; i++)
```

```
    {
```

```
        scanf("%d", &a[i]);
```

```
    }
```



```
for (i=0; i<n; i++)
```

```
{ if (a[i] % 2 == 0)
```

```
    even++;
```

```
    else
```

```
        odd++;
```

```
}
```

```
printf ("even elements = %d \n", even);
```

```
printf ("odd elements = %d \n", odd);
```

```
return 0;
```

```
}
```

output:

enter number of elements = 5

enter array of elements = 7 12 13 18 32

even elements = 3

odd elements = 2

2. Write a c program to ~~count no~~ print fibonacci series and sum of fibonacci series.

a. Aim:

To write a c program to print fibonacci series and sum of fibonacci series.

Algorithm:

1. Start

2. Read the number of terms

3. Initialize a=0 and b=1, sum=0.

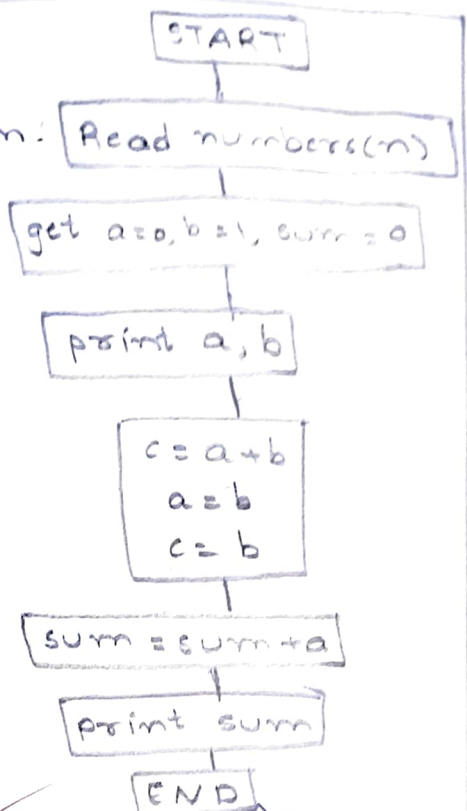
5. Print fibonacci term.
6. Add each term to sum.
7. Display the total sum
8. stop

Program :

```
#include <stdio.h>
int main ( )
{
    int n, i;
    int a=0, b=1, c;
    int sum=0;
    printf("enter number of terms: ");
    scanf ("%d", &n);
    printf ("fibonacci series: \n");
    for (i=1; i<=n; i++)
    {
        printf ("%d", a);
        sum = sum + a;
        c = a + b;
        a = b;
        b = c;
    }
    printf ("sum: %d", sum);
    return 0;
}
```

output:

enter no-of elements : 5
 fibonacci series: 0 1 1 2 3 sum = 7



3. Write the concept map for data structures and its classification.

a.

